Module 03

Installing OpenClaw 🦞

From Zero to a Running AI Agent

🧭 Navigation Chart

By the end of this module, you will be able to:

1. **Run** the 🦞 OpenClaw installer in your Ubuntu terminal
2. **Walk through** every step of the onboarding wizard
3. **Choose** an AI provider and configure your 🔑 API key
4. **Avoid** the token formatting gotcha that breaks most first installs
5. **Configure** the ⛵ gateway for secure local operation
6. **Install** the daemon so OpenClaw runs 24/7
7. **Verify** that everything is working
8. **Access** the dashboard and TUI (Terminal User Interface)

📜 Ship's Logbook

Term	Definition
⛵ Gateway	The core OpenClaw process -- coordinates messages, AI models, and actions
Onboarding wizard	Interactive setup process after installation
🔑 API key	Secret credential letting OpenClaw talk to your AI provider
TUI	Terminal User Interface -- text-based chat in your terminal
Control UI	Web-based dashboard for managing OpenClaw
Provider	Company whose AI model you use (Anthropic, OpenAI, Google, etc.)
Daemon	Background service that keeps the gateway running 24/7
Token	(1) Authentication credential, or (2) units of text the AI processes

Before You Start

Make sure you have completed Module 02:

- [ ] Ubuntu installed and running
- [ ] systemd running (out of the box on native Ubuntu)
- [ ] Node.js 22+ installed
- [ ] Your Ubuntu terminal open

Everything in this module happens in the Ubuntu terminal.

Also decide which AI provider you want to use...

AI Provider Comparison

Provider	Model	Cost	Free?
Google	Gemini 2.5 Flash	Free	Yes — ~250 req/day
Anthropic	Claude Opus 4.6	~$200/mo	No
Anthropic	Claude Sonnet 4.5	~$50-100/mo	No
OpenAI	GPT-4 / GPT-5.2 Turbo	~$20-200/mo	No
Kimi	K2.5	Very cheap	No
MiniMax	M2.5	~$10/mo	No

Two paths: *Free (Gemini) or Recommended (Anthropic Claude).*

Choose Your Path

Path A: Start Free — Google Gemini 2.5 Flash

- No credit card required — just your Google account
- ~250 free requests/day — plenty for learning
- Perfect for: "I want to try this before I spend a dollar"

Path B: Full Power — Anthropic Claude

- Best quality, strongest prompt injection resistance
- Costs ~$20-200/month
- Perfect for: "I want the best experience from day one"

> ***Not sure?*** *Start free. You can switch anytime (Module 09).*

🚩 Rough Waters: Claude Max/Pro Subscriptions

Path A (Gemini) users: these next two warnings are for paid providers. Listen anyway — you'll need this when you upgrade.

Can you use your Claude Max or Pro subscription with 🦞 OpenClaw? No.

- Using Max/Pro with OpenClaw is **against Anthropic's Terms of Service**
- People have been **banned** -- losing access to **all Anthropic products**
- Use the **Anthropic API** with a separate 🔑 API key (pay-as-you-go)
- Set a **spending limit** ($20-50/month) — Module 09 covers reducing costs by **90%**

🚩 Rough Waters: No Spending Limits

Real story: *One user ran 8 agents on pay-per-use API without a spending limit.* ***$800 burned in less than a week.***

Protection	How	Priority
Hard spending limit	Anthropic Console > Settings > Usage Limits > $20-50/month	Non-negotiable
Start with one agent	Understand your token burn rate first	Critical
Monitor daily (first week)	Check console.anthropic.com/usage every day	High
Use cheap models for routine	Haiku for heartbeats, Sonnet for simple tasks (Module 09)	High

Step 1: Install 🦞 OpenClaw

In your 🐚 Ubuntu terminal, run:

```
curl -fsSL https://openclaw.ai/install.sh | bash
```

What this does:

- Downloads the install script from openclaw.ai
- `-fsSL` = fail silently, no progress bar, follow redirects
- Pipes the script into bash to execute

Step 1: Install 🦞 OpenClaw (continued)

Expected output:

```
Installing OpenClaw...
Checking Node.js version... v22.x.x
Installing OpenClaw packages...
Installation complete!
```

The **onboarding wizard starts automatically** after install. When asked "Quick Setup" or "Manual" -- choose **Manual**.

The wizard asks a series of questions. We will walk through **every single one**.

> *Take your time. If you make a mistake, you can re-run the wizard with* `npx openclaw config`.

Wizard Q1-Q2: ⛵ Gateway Type and Workspace

Question 1: Gateway Type

```
? Gateway type: > Local Gateway
```

Choose: Local Gateway -- runs on this machine.

Question 2: Workspace Directory

```
? Workspace directory: (~/.openclaw)
```

Accept the default (press Enter). Creates `~/.openclaw/` in your home directory.

Wizard Q3: AI Model Provider

```
? Select your AI provider:
  > Anthropic
    OpenAI
    Google
    MiniMax
    Open Router
    Custom
```

Path A (Free): Choose **Google**

Path B (Full Power): Choose **Anthropic**

Wizard Q4: 🔑 API Key Setup

Path A (Free — Gemini):

1. Go to **aistudio.google.com** — sign in with your Google account
2. Click **Get API Key** (bottom left) — a key already exists in your default project
3. Copy the key — no billing or spending limit needed

Path B (Paid — Anthropic):

1. Go to **console.anthropic.com** — create account, verify email
2. **Settings > Billing** — add payment, **set spending limit** ($20-50/month)
3. **Settings > API Keys** → "Create Key" → name it "OpenClaw"
4. Copy the key (starts with `sk-ant-...`) — you can only see it once

Both paths: use the **text editor trick** on the next slide before pasting.

🚩 Rough Waters: The Token Formatting Gotcha

The Problem (the #1 install error): Browsers add hidden characters (line breaks, spaces) when copying. Pasting directly into the terminal **corrupts the key**.

The Solution — The Text Editor Trick:

1. Copy the 🔑 API key from your provider's console
2. Paste into a **text editor** (Text Editor app or `gedit`) -- verify it is **one single line**
3. Select all (`Ctrl + A`), copy (`Ctrl + C`)
4. Paste into the terminal (`Ctrl + Shift + V`)

```
? Enter your API key: xxxxxxxxxxxxxxxxxxxxxxxxxxxxxxxx
```

Wizard Q5-Q6: Model and Port

Question 5: Model Selection

Path A (Gemini): Choose `gemini-2.5-flash` — covered by the free tier

Path B (Anthropic): Choose `claude-opus-4-6` — most capable model

Question 6: ⛵ Gateway Port

```
? Gateway port: (18789)
```

Accept the default (press Enter). Port 18789 is the standard OpenClaw port.

Wizard Q7: ⛵ Gateway Bind

```
? Gateway bind: > Loopback
```

Choose: Loopback (most secure -- only this computer can connect)

Option	Who Can Connect	When to Use
Loopback	Only this computer	Default -- most secure
LAN	Devices on your WiFi	Access from another local machine
Tailscale	Your Tailscale network	Secure remote access
Custom	Whatever IP you specify	Advanced users only

> *Change later with* `npx openclaw config` *(re-runs the wizard)*

Wizard Q8-Q9: Tailscale and Token

Question 8: Tailscale Exposure

```
? Tailscale exposure: (Off)
```

Choose: Off -- enable later when security is fully configured.

Question 9: ⛵ Gateway Token

```
? Gateway token: (leave blank to auto-generate)
```

Leave blank -- auto-generates a secure random token. **Save it when displayed.**

Wizard Q10: Chat Channels

```
? Configure chat channels? > Skip for now
```

Choose: Skip for now -- we set up Telegram in Module 06.

Get the base installation working first. You can always add channels later with `npx openclaw config` (re-runs the wizard).

Wizard Q11-Q14: 🐠 Skills, Hooks, Service

- **Q11: Skills** -- Skip all (covered in Module 07)
- **Q12: Hooks** -- Enable BOOT.md and session 🪸 memory
- **Q13: Service Runtime** -- Choose Node.js (only option)
- **Q14: Install as Background Service** -- Choose **Y**

If the wizard says "Gateway service already installed" -- choose **Restart**

When asked "How do you want to hatch your bot?" -- choose **do this later**

When asked "Enable bash shell completion?" -- choose **Yes**

Step 2: Onboarding Complete

The wizard finishes with output like this:

```
Gateway configured
AI provider connected (your chosen provider and model)
Daemon installed (systemd service)
Dashboard:     http://127.0.0.1:18789/
Gateway Token: abc123xyz... (save this!)
```

Save your gateway token immediately -- store it in a password manager or secure note. You need it for dashboard access.

Retrieve it later with: `cat ~/.openclaw/openclaw.json` (look for the token field)

Step 2b: Approve Device Pairing

After onboarding, the TUI requires **device pairing approval** before it can connect.

If you see "pairing required" when launching the TUI:

```
npx openclaw devices list --json
```

Find the `requestId` for the pending entry, then approve it:

```
npx openclaw devices approve YOUR-REQUEST-ID
```

Replace `YOUR-REQUEST-ID` with the actual ID from the list output.

Immediately After Onboarding: Install QMD

Before doing anything else, install **QMD** -- an on-device search engine for your agent's 🪸 memory files:

```
npx clawhub install qmd
npx openclaw gateway restart
```

(If npx says "Need to install clawhub" -- type **y**. ClawHub is the skill marketplace CLI.)

Why this matters:

- Without QMD, your agent re-reads files every time it needs information (wastes tokens)
- With QMD, your agent can search indexed files instantly
- Saves money from day one by avoiding redundant file reads
- Takes 30 seconds to install

Step 3: Verify the Installation

Run these three commands in order:

```
npx openclaw status                      # Should show: Running
npx openclaw doctor                      # All critical checks should pass
npx openclaw security audit --deep --fix # Fixes file permissions
```

- If `doctor` reports failures, run `npx openclaw doctor --repair`
- Security audit sets dirs to 700 and files to 600

Step 4: Access the Dashboard

1. In your terminal, run: `npx openclaw dashboard --no-open`
2. Copy the **full URL** it prints (includes your token)
3. Paste it into your browser

> *If the dashboard won't connect, skip it — the **TUI is the primary interface**.*

Dashboard vs. TUI:

Interface	What It Is	Best For
Control UI (Dashboard)	Web page at localhost:18789	Configuration, monitoring, logs
TUI (Terminal UI)	Text chat in your terminal	Talking to your agent, running commands

Most day-to-day interaction happens in the **TUI**. The dashboard is for **management**.

Step 5: Your First Chat

```
npx openclaw tui
```

A full-screen terminal interface opens. Type a simple greeting:

```
Hello! My name is [your name]. What's your name?
```

Step 5: Your First Chat (continued)

What should happen:

- Your message appears in the chat
- After 1-5 seconds, the AI responds
- The token counter increments

If you get a response -- 🦞 OpenClaw is working!

> *Do not have a full conversation yet. We will do that in Module 04. Type* `/exit` *or* `Ctrl + C` *to leave.*

Step 6: Verify the Daemon

Check the service status

```
npx openclaw gateway status    # Look for: Active: active (running)
```

Test terminal independence

1. **Close** your Ubuntu terminal completely
2. Wait 10 seconds, then open `http://127.0.0.1:18789/`
3. Dashboard should **still load** -- the daemon kept it alive

Step 6: Service Commands Reference

```
npx openclaw gateway start     # Start the daemon
npx openclaw gateway stop      # Stop the daemon
npx openclaw gateway restart   # Restart the daemon
npx openclaw logs      # View daemon logs
```

Use `restart` after any config change. Use `logs` to debug startup issues.

🚩 Damage Control

Problem	Fix
`openclaw: command not found`	Use `npx openclaw` instead -- the install does not add it to PATH
"API key invalid"	Use the **text editor trick** -- hidden line breaks are the #1 cause
"Port already in use"	`npx openclaw config` (re-run wizard, change port), then restart
"Connection refused" in browser	Check `npx openclaw gateway status`; use `http://` not `https://`
Daemon does not start on boot	Verify systemd: `systemctl is-system-running`
AI responses very slow (30s+)	Check `status.anthropic.com`; try Sonnet 4.5 temporarily

🚩 Damage Control (continued)

Problem	Fix
"Insufficient credits"	Check billing at your provider's console
Token counter climbs fast	Use `/compact` to compress history; start new chats with `/new`
"pairing required"	Run `npx openclaw devices list --json`, find requestId, approve with `npx openclaw devices approve ID`
"auth failed" on dashboard	Run `npx openclaw dashboard --no-open` to get a tokenized URL. Paste the full URL into your browser.
Dashboard still won't connect	Skip it — the TUI is the primary interface
`http` vs `https`	Use `http://` not `https://` for the dashboard URL

⚙️ Hands on Deck

Complete this checklist before proceeding to Module 04:

- [] `npx openclaw --version` shows a version number
- [] `npx openclaw status` shows "Running" and `npx openclaw doctor` passes
- [] `npx openclaw security audit --deep` has no critical issues
- [] Dashboard loads at `http://127.0.0.1:18789/`
- [] TUI opens with `npx openclaw tui` and AI responds to a test message
- [] ⛵ Gateway stays running after closing the terminal
- [] Gateway token saved somewhere secure
- [] `~/.openclaw/` directory is mode 700

Bonus: Run `npx openclaw logs` and read the last few lines.

💎 Treasure Chest

1. **Installation is one command** -- the onboarding wizard starts automatically
2. **The token formatting gotcha is real** -- always paste API keys into a text editor first
3. **Start with loopback binding** -- most secure, only this machine can connect
4. **Set API spending limits** (paid providers) -- Gemini free users skip this step
5. **The daemon keeps the gateway running 24/7** -- even after closing the terminal
6. **Two interfaces**: TUI for chatting, Dashboard for managing
7. **Run the security audit** after installation -- catches permission issues
8. **Save your gateway token** -- retrieve with `cat ~/.openclaw/openclaw.json`
9. **Everything stays local except API calls** -- your data never leaves your machine

🌊 Next Port of Call

Module 04: Your First Conversation 🦞

> *Your agent is awake. Time to introduce yourself, do a "brain dump" of your life and goals, and learn the essential commands for daily use.*